

		on a smooth surface. Hence the centre of gravity of the system does not move.
6	B	Conservation of momentum: $mu = mv + m(2v) \rightarrow u = 3v$ Final kinetic energy of system, $K_f = \frac{1}{2}mv^2 + \frac{1}{2}m(2v)^2 = \frac{1}{2}m(5)v^2$ Initial kinetic energy of system, $K_i = \frac{1}{2}mu^2 = \frac{1}{2}m(3v)^2 = \frac{1}{2}m(9)v^2$ Loss in kinetic energy of system $= K_i - K_f = \frac{1}{2}m(9 - 5)v^2 = \frac{1}{2}m(4)v^2$ Fractional loss in kinetic energy $= \frac{\frac{1}{2}m(4)v^2}{\frac{1}{2}m(9)v^2} = \frac{4}{9}$
7	A	To obey Hooke's Law, the given F-x graph should be linear. Work done, $W = \text{area under } F \text{ vs } x \text{ graph (for linear part of the graph)}$